# Evaluation runs: rpc (rpc-at-most-once-aneris)

generated 2026-11-05 from brain-data bb4df53ad 2026-11-05 15:31:21 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

No run of the stateless engine on this benchmark.

## stateful

No run of the stateful engine on this benchmark.

## offline

### offline: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 6 | 146 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 20 | 1070 | conforms | 0.1 | 4 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 70 | 6622 | conforms | 0.5 | 4 | 10 min, 16 GB | 0.1 | 0.5 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 252 | 37212 | conforms | 1.9 | 14 | 10 min, 16 GB | 0.2 | 1.8 | 0.1 | in-memory | brain05, 2026-11-02 |
| 6 | 924 | 196548 | conforms | 5.0 | 62 | 10 min, 16 GB | 0.1 | 4.5 | 0.4 | in-memory | brain05, 2026-11-02 |
| 7 | 3432 | 994422 | conforms | 21.3 | 291 | 10 min, 16 GB | 0.3 | 19.4 | 1.9 | in-memory | brain05, 2026-11-02 |
| 8 | 12870 | 4874870 | conforms | 107.6 | 1352 | 10 min, 16 GB | 0.8 | 98.0 | 9.6 | in-memory | brain05, 2026-11-02 |
| 9 | 48620 | 23327876 | conforms | 666.8 | 6994 | 10 min, 16 GB | 1.8 | 614.9 | 51.9 | in-memory | brain05, 2026-11-02 |

### offline: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 0.2 | 4 | 10 min, 16 GB | 0.2 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 2 | 6 | 166 | violation | 0.0 | 4 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 20 | 1350 | violation | 0.2 | 4 | 10 min, 16 GB | 0.1 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 70 | 9142 | violation | 0.6 | 6 | 10 min, 16 GB | 0.1 | 0.5 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 252 | 55692 | violation | 2.1 | 22 | 10 min, 16 GB | 0.2 | 2.0 | 0.1 | in-memory | brain05, 2026-11-02 |
| 6 | 924 | 316668 | violation | 6.5 | 110 | 10 min, 16 GB | 0.1 | 5.8 | 0.7 | in-memory | brain05, 2026-11-02 |
| 7 | 3432 | 1715142 | violation | 34.1 | 506 | 10 min, 16 GB | 0.3 | 30.8 | 3.3 | in-memory | brain05, 2026-11-02 |
| 8 | 12870 | 8958950 | violation | 205.9 | 2568 | 10 min, 16 GB | 0.8 | 187.1 | 18.8 | in-memory | brain05, 2026-11-02 |
| 9 | 48620 | 45498596 | violation | 1409.6 | 12761 | 10 min, 16 GB | 1.8 | 1321.5 | 88.0 | in-memory | brain05, 2026-11-02 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 12 dedup bag, 12 no-dedup bag.
skipped: a smaller size had reached its limit: 10 no-dedup bag, 10 dedup bag.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 3.2 | 164 | 10 min, 16 GB | 0.2 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Java implementation against the Rust specification: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 3.2 | 173 | 10 min, 16 GB | 0.2 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |

**Not run (cross offline, Java implementation against the Rust specification).** refused (an input missing at run time): 2 bag dedup, 2 bag no-dedup, 3 bag dedup, 3 bag no-dedup, 4 bag no-dedup, 4 bag dedup.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 2.9 | 163 | 10 min, 16 GB | 2.9 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 146 | conforms | 3.1 | 157 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 1070 | conforms | 3.1 | 169 | 10 min, 16 GB | 3.1 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 70 | 6622 | conforms | 3.5 | 176 | 10 min, 16 GB | 3.5 | 0.5 | 0.0 | in-memory | brain01, 2026-11-02 |

### cross offline, Rust implementation against the Java specification: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 2.9 | 163 | 10 min, 16 GB | 2.9 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 6 | 166 | violation | 3.1 | 157 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 20 | 1350 | violation | 3.1 | 169 | 10 min, 16 GB | 3.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 70 | 9142 | violation | 3.5 | 176 | 10 min, 16 GB | 3.5 | 0.5 | 0.0 | in-memory | brain01, 2026-11-02 |

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 6 | 146 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 20 | 1070 | conforms | 0.2 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 70 | 6622 | conforms | 0.7 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 14 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 2 | 6 | 166 | violation | 0.0 | 4 | 1 h, 250 GB | 3 | brain01, 2026-11-02 |
| 3 | 20 | 1350 | violation | 0.2 | 4 | 1 h, 250 GB | 7 | brain01, 2026-11-02 |
| 4 | 70 | 9142 | violation | 1.0 | 4 | 1 h, 250 GB | 31 | brain01, 2026-11-02 |

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.2 | 164 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 146 | 146 | conforms | 3.8 | 163 | 10 min, 16 GB | 3.8 | 0.1 | 0.1 | in-memory | brain01, 2026-11-02 |
| 3 | 1070 | 1070 | conforms | 8.0 | 782 | 10 min, 16 GB | 7.9 | 0.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 6622 | 6622 | conforms | 44.8 | 6394 | 10 min, 16 GB | 44.8 | 0.5 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Rust implementation against its Java twin: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.2 | 173 | 10 min, 16 GB | 3.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 166 | 166 | conforms | 4.0 | 172 | 10 min, 16 GB | 4.0 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 1350 | 1350 | conforms | 8.5 | 783 | 10 min, 16 GB | 8.5 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 9142 | 9142 | conforms | 53.2 | 7009 | 10 min, 16 GB | 53.2 | 0.5 | 0.0 | in-memory | brain01, 2026-11-02 |

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.1 | 164 | 10 min, 16 GB | 0.0 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 146 | 146 | conforms | 3.8 | 163 | 10 min, 16 GB | 0.1 | 3.8 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 1070 | 1070 | conforms | 7.9 | 782 | 10 min, 16 GB | 0.1 | 7.9 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 6622 | 6622 | conforms | 44.8 | 6394 | 10 min, 16 GB | 0.5 | 44.8 | 0.0 | in-memory | brain01, 2026-11-02 |

### twins offline, Java implementation against its Rust twin: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.1 | 173 | 10 min, 16 GB | 0.2 | 3.1 | 0.0 | in-memory | brain01, 2026-11-02 |
| 2 | 166 | 166 | conforms | 4.0 | 172 | 10 min, 16 GB | 0.0 | 4.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 1350 | 1350 | conforms | 8.5 | 783 | 10 min, 16 GB | 0.2 | 8.5 | 0.0 | in-memory | brain01, 2026-11-02 |
| 4 | 9142 | 9142 | conforms | 53.2 | 7009 | 10 min, 16 GB | 0.5 | 53.2 | 0.0 | in-memory | brain01, 2026-11-02 |

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.3 | 157 | 10 min, 16 GB | 3.3 | 0.8 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 146 | 146 | conforms | 4.5 | 154 | 10 min, 16 GB | 4.4 | 0.2 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3 | 1070 | 1070 | conforms | 8.7 | 754 | 10 min, 16 GB | 8.5 | 0.7 | 0.2 | in-memory | brain02, 2026-11-03 |
| 4 | 6622 | 6622 | conforms | 40.4 | 6734 | 10 min, 16 GB | 39.7 | 0.9 | 0.6 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.3 | 160 | 10 min, 16 GB | 3.3 | 0.4 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 166 | 166 | conforms | 4.2 | 156 | 10 min, 16 GB | 4.2 | 0.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 1350 | 1350 | conforms | 8.7 | 809 | 10 min, 16 GB | 8.7 | 0.6 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 9142 | 9142 | conforms | 50.9 | 7308 | 10 min, 16 GB | 50.9 | 0.7 | 0.1 | in-memory | brain02, 2026-11-03 |

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.3 | 157 | 10 min, 16 GB | 0.8 | 3.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 146 | 146 | conforms | 4.5 | 154 | 10 min, 16 GB | 0.2 | 4.4 | 0.1 | in-memory | brain02, 2026-11-03 |
| 3 | 1070 | 1070 | conforms | 8.7 | 754 | 10 min, 16 GB | 0.7 | 8.5 | 0.2 | in-memory | brain02, 2026-11-03 |
| 4 | 6622 | 6622 | conforms | 40.4 | 6734 | 10 min, 16 GB | 0.9 | 39.7 | 0.6 | in-memory | brain02, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.3 | 160 | 10 min, 16 GB | 0.4 | 3.3 | 0.0 | in-memory | brain02, 2026-11-03 |
| 2 | 166 | 166 | conforms | 4.2 | 156 | 10 min, 16 GB | 0.2 | 4.2 | 0.0 | in-memory | brain02, 2026-11-03 |
| 3 | 1350 | 1350 | conforms | 8.7 | 809 | 10 min, 16 GB | 0.6 | 8.7 | 0.0 | in-memory | brain02, 2026-11-03 |
| 4 | 9142 | 9142 | conforms | 50.9 | 7308 | 10 min, 16 GB | 0.7 | 50.9 | 0.1 | in-memory | brain02, 2026-11-03 |

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 3.8 | 167 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 146 | 146 | conforms | 4.2 | 167 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 1070 | 1070 | conforms | 9.7 | 778 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 6622 | 6622 | conforms | 41.1 | 6770 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 4.0 | 166 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 166 | 166 | conforms | 4.2 | 161 | 1 h, 250 GB | | brain02, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 3 | 1350 | 1350 | conforms | 9.6 | 765 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 9142 | 9142 | conforms | 53.7 | 7249 | 1 h, 250 GB | | brain02, 2026-11-03 |

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 0.9 | 2 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 146 | 146 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 1070 | 1070 | conforms | 0.2 | 2 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 6622 | 6622 | conforms | 0.8 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-dedup**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1 | 14 | 14 | conforms | 0.0 | 2 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 2 | 166 | 166 | conforms | 0.0 | 4 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 3 | 1350 | 1350 | conforms | 0.2 | 2 | 1 h, 250 GB | | brain02, 2026-11-03 |
| 4 | 9142 | 9142 | conforms | 0.7 | 8 | 1 h, 250 GB | | brain02, 2026-11-03 |